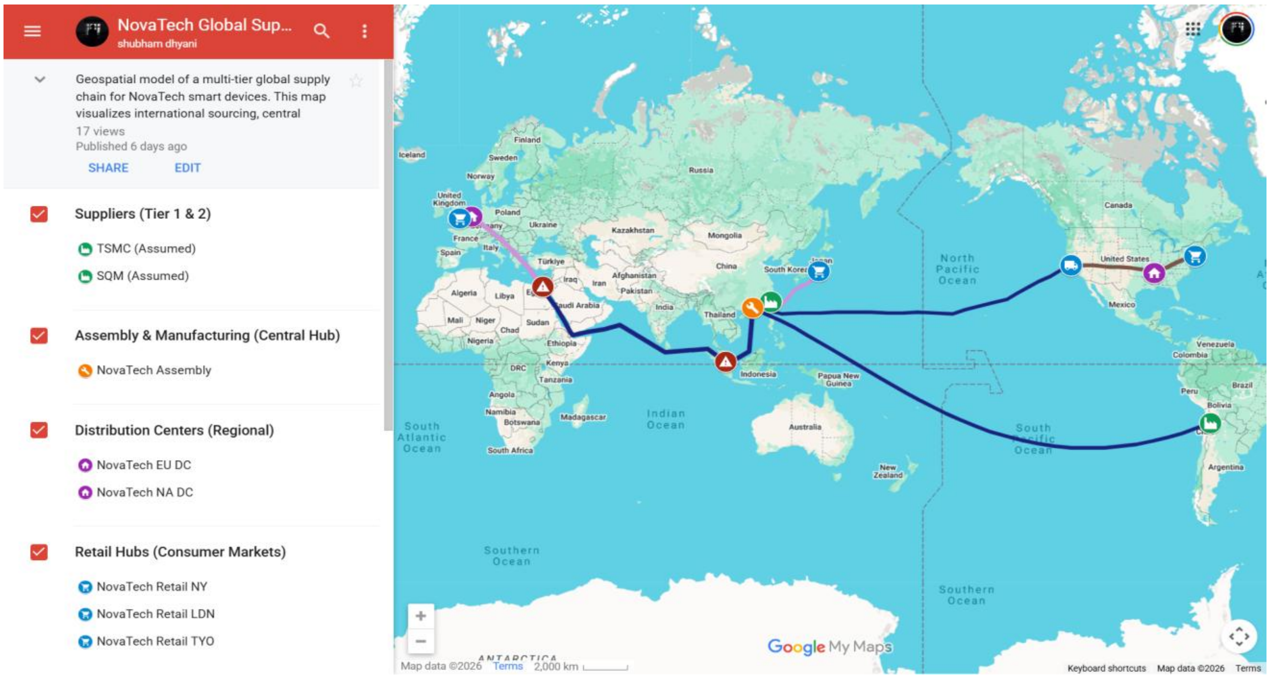
Global Supply Chain Mapping & Network Design

1. Geospatial Map & Network Link



Interactive Google MyMaps Link:
https://www.google.com/maps/d/viewer?mid=1KSgWZIoUJ88-P_7eQMH3DztryO9eR0c&ll=34.2319466350072%2C146.6015625&z=2

2. AI Pedagogy & Tools Used

In accordance with the Green Light AI Policy for this lab, the following tools were utilized for brainstorming and research:

Tools Used:

- Gemini (AI Research Assistant): Used to estimate structural cost differences, brainstorm Tier 1 supplier locations, and evaluate maritime disruption scenarios.
- Google MyMaps: Used for geospatial plotting, route drawing, and risk node mapping.

Prompts Used:

- "Act as a supply chain consultant. Suggest realistic Tier 1 global suppliers for raw lithium and microchips for a high-end smart device, comparing structural costs between alternative countries."
- "Analyze the transit times, cost trade-offs, and transportation modes (ocean vs. multimodal) for moving electronics from Shenzhen to European and US markets."
- "Identify major maritime choke points between East Asia and Europe/US East Coast, and explain the strategic risk to customer responsiveness if a disruption occurs."

3. Product & Company Overview

Company: NovaTech Global
Product: NovaTech ProXR Smart Glasses
Rationale: To fulfill the requirement of designing a supply chain for a "high-end smart device," this product requires advanced microprocessors, lightweight battery technology, and high-quality optical lenses, making a complex, multi-continent global supply chain essential.

4. Network Node Identification (Phase 1)

Node Type	Location	Role / Justification
Tier 1 (Supplier A)	Hsinchu, Taiwan	Sources advanced microchips (semiconductors).
Tier 1 (Supplier B)	Antofagasta, Chile	Sources raw lithium for high-density batteries.
Tier 2 (Assembly Plant)	Shenzhen, China	The primary assembly plant.
Tier 3 (DC 1)	Southern Europe / Rotterdam	Serves the European market via transshipment.
Tier 3 (DC 2)	Los Angeles, USA	Serves the North American market.
Tier 4 (Retail Hub 1)	Berlin/UK, Europe	Major tech-consumer end-market.
Tier 4 (Retail Hub 2)	New York City, USA	High-density East Coast retail hub.
Tier 4 (Retail Hub 3)	Tokyo, Japan	High-income Asian end-consumer market.

5. Logistics Routes (Phase 2)

- Route 1 (Taiwan to Shenzhen): Yellow (Air Freight). Minimizes transit time for high-value chips.
- Route 2 (Chile to Shenzhen): Blue (Ocean Freight). Keeps transportation costs low for heavy, bulk lithium.
- Route 3 (Shenzhen to Europe): Blue to Purple Split (Ocean to Multimodal). Ocean freight to Southern Europe, shifting to rail/truck for final delivery to balance cost and speed.
- Route 4 (Shenzhen to Japan): Blue to Purple Split (Ocean to Multimodal). Short-sea shipping transferring to domestic Japanese rail.
- Route 5 (Shenzhen to US East Coast): Blue to Brown Split (Ocean to Land). Ocean freight to LA, transferring to transcontinental rail/trucking.

6. Comparative Sourcing Strategies (R&D)

- Lithium (Chile vs. Australia): Chile utilizes cost-effective brine extraction ($3k-$5k/ton) over Australia's expensive hard-rock mining.
- Microchips (Taiwan vs. US/China): Taiwan ensures leading-edge tech without the operational premiums of the US or the legacy-node technology gap of China.
- Assembly (Shenzhen vs. India): Shenzhen wins on ecosystem density and productivity, offsetting higher wage rates.

7. Risk Identification & Choke Points

- Strait of Malacca (Risk Node 1): Highly congested shipping lane connecting East Asia to the Indian Ocean.
- Suez Canal (Risk Node 2): Critical bottleneck connecting the Red Sea to the Mediterranean.

8. Risk Mitigation Strategies

Operating a decentralized global supply chain exposes the network to various external and internal vulnerabilities. To ensure operational resilience and maintain service level agreements across all international nodes, NovaTech must implement proactive mitigation strategies across five primary risk categories.

1. Strategic Stockpiling and Inventory Buffers

Strict Just-in-Time (JIT) manufacturing minimizes holding costs but leaves the network highly exposed to sudden supply shocks, where a single missing component can halt entire assembly lines.

- **Mitigation:** Transition to a hybrid "Just-in-Case" model by maintaining strategic reserves of critical, difficult-to-source materials (such as proprietary semiconductors or rare earth elements). Calibrate dynamic safety stock levels at regional distribution centers to act as shock absorbers, ensuring localized disruptions do not immediately impact final consumer fulfillment.

2. Geopolitical and Trade Policy Risks

Heavy reliance on specific global regions introduces vulnerability to trade embargoes, tariff fluctuations, and political instability.

- **Mitigation:** Implement a "China +1" or nearshoring strategy to diversify manufacturing and assembly nodes. Continuously monitor regional geopolitical climates to anticipate trade policy shifts, allowing for the preemptive rerouting of critical component sourcing before bottlenecks occur.

3. Transportation and Logistics Disruptions

Relying on single modes of transit leaves the network vulnerable to maritime chokepoint closures, port strikes, or sudden capacity shortages.

- **Mitigation:** Establish multi-modal transportation redundancies. While standard tech components may utilize cost-effective ocean freight, establish pre-negotiated contracts for air freight to serve as an immediate backup for high-margin or critical components. Utilize real-time geographic tracking to enable dynamic rerouting when primary transit lanes are compromised.

4. Supplier and Node Dependency

Over-reliance on a single Tier-1 supplier or a single geographic manufacturing node creates a single point of failure.

- **Mitigation:** Develop a comprehensive Tier-1 and Tier-2 supplier map to identify hidden dependencies. Maintain a diversified supplier base and avoid concentrating all high-value manufacturing in a single facility.

5. Demand Volatility and Forecasting Errors

Sudden shifts in consumer technology trends can lead to severe stockouts or excess obsolete inventory.

- **Mitigation:** Leverage predictive analytics and machine learning models to transition from reactive to proactive demand forecasting. Implement an agile inventory management system that allows for rapid reallocation of stock between regional distribution nodes based on real-time market signals.

Summary Matrix for Network Resilience

Risk Category	Primary Vulnerability	Strategic Mitigation
Inventory	JIT vulnerability to sudden supply shocks	Strategic stockpiling, dynamic safety stock at regional hubs
Geopolitical	Tariffs, regional instability, trade wars	Geographic diversification of nodes, nearshoring
Logistics	Chokepoint delays, port closures	Multi-modal redundancies, dynamic routing
Supply	Single point of failure at the supplier level	Multi-sourcing, supplier mapping
Demand	Bullwhip effect, forecast inaccuracies	AI-driven predictive forecasting, agile inventory

9. Strategic Reflection (Phase 3 Questions)

Q1: How do the distances and transportation modes you selected impact the total cost of your global sourcing decisions?

Answer: The vast geographic distances required to source raw lithium (Chile) and advanced chips (Taiwan) inherently increase supply chain lead times and pipeline inventory. To optimize the total landed cost, I selected transportation modes based on product density and value. Low-weight, high-value components (chips) utilize costly air freight to prevent massive capital tie-up. Conversely, heavy materials and finished goods utilize cost-effective ocean freight. By implementing transshipment routing (e.g., shifting from ocean to multimodal rail in Europe and the US), the network offsets the long ocean transit times, bridging the gap between cost-efficiency and customer responsiveness.

Q2: If a disruption occurs at your identified choke point, how will it impact your network's strategic fit and customer responsiveness?

Answer: A disruption at either the Strait of Malacca or the Suez Canal would severely break the network's strategic fit. The strategic fit of NovaTech relies on balancing low ocean transport costs with acceptable lead times. If a blockage occurs, ships must reroute around the Cape of Good Hope (adding 10-15 days). This destroys customer responsiveness, leading to stockouts in European retail hubs. To regain that responsiveness and prevent stockouts, NovaTech would be forced to panic-ship inventory via Air Freight, completely erasing the product's profit margins.